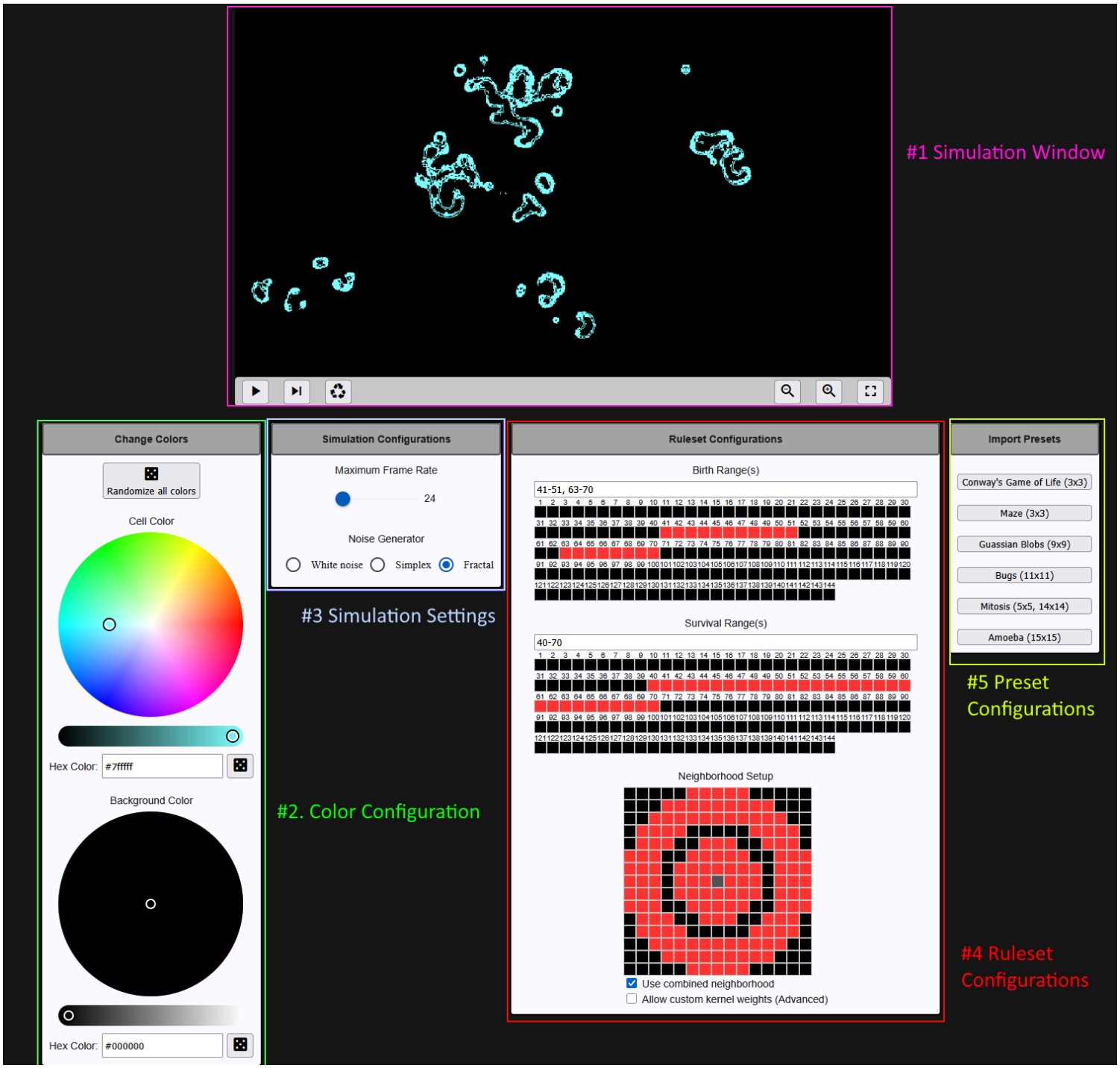
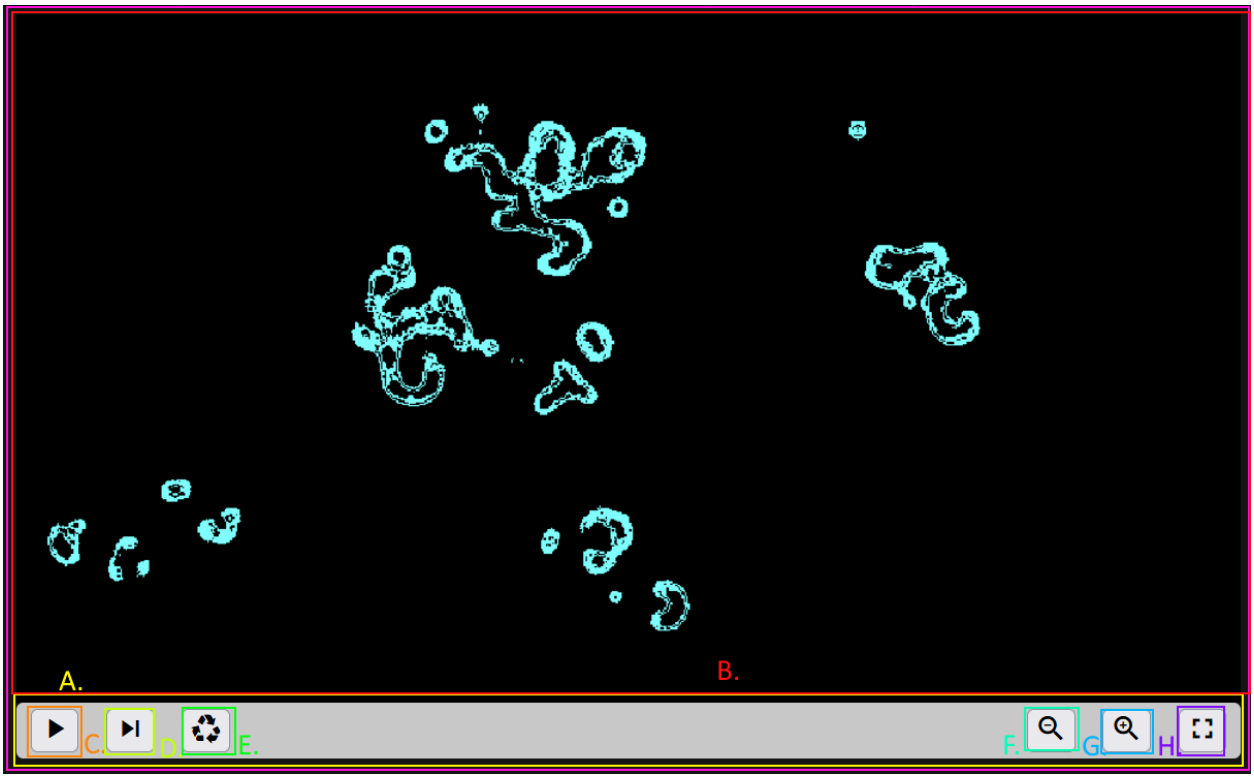
Cytomachia UI specification

Full UI, zoomed out:



#1 Simulation Window:



A. Simulation Bar

- Description: bar at the bottom of the simulation window for holding buttons responsible for changing the simulation state.

B. Simulation Window

- Description: Canvas window for displaying the current camera view of the current simulation state.
- Interaction:
  - **i.** Hover: Cursor shows grab/grabbing cursor
  - **ii.** Scroll: Scrolling up/down zooms the camera in/out at the cursor's location
  - **iii.** Click+drag: Clicking and dragging the canvas will move the camera along with the mouse
  - **iv.** Arrow Keys: Pressing/Holding arrow keys will move camera based on the direction of the arrow key pressed.
- Visual Changes:
  - **i.** Updates along with the render loop to render camera view each animation frame

C. **Pause/Play Buton**
    - Description: Button for toggling pausing/unpausing the simulation from performing its compute pass
    - Visual Changes:
        i. Hover: Highlight, description tooltip
        ii. Symbol changes based on if simulation is paused or running
    - Interaction:
        i. Click: Toggles between the paused/unpaused state
        ii. Spacebar: Toggles between the pause/unpaused state
D. **Pause/Play Buton**
    - Description: Button stepping the simulation once
    - Interaction:
        i. Hover: Highlight, description tooltip
        ii. Click: Steps simulation once
        iii. N-key: Steps simulation once
E. **Reseed simulation button**
    - Description: Button for reseeding simulation
    - Interaction:
        i. Hover: Highlight, description tooltip
        ii. Click: Reseeds simulation with noise generator
        iii. R-key: Reseeds simulation with noise generator
F. **Zoom out button**
    - Description: Button for zooming out
    - Interaction:
        i. Hover: Highlight, description tooltip
        ii. Click: Zooms the camera out
G. **Zoom in button**
    - Description: Button for zooming in
    - Interaction:
        i. Hover: Highlight, description tooltip
        ii. Click: Zooms the camera in
H. **Expand view button**
    - Description: Button for toggling the expanded view mode
    - Interaction:
        i. Hover: Highlight, description tooltip
        ii. Click: Toggles expanded view for simulation window
        iii. Expanded view allows simulation window to take up full screen

#2 Color configuration

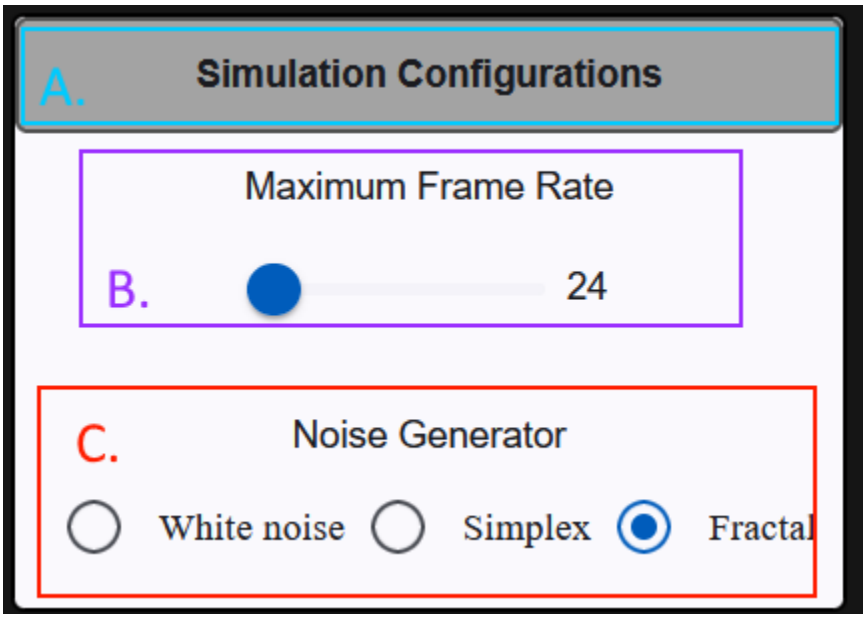
A. **Title**
    - Title of configuration group
B. **Randomize All Colors button**
    - Description: Button for randomizing the colors of both the cell and the background
    - Interaction:
        i. Hover: Highlight, description tooltip
        ii. Click: Randomly assign color to both the cell and background
C. **Cell color Hue/Saturation picker**
    - Description: Wheel for picking cell color hue and saturation
    - Interaction:
        i. DragEnd: Update cell color to new value and update hex color input
D. **Cell color light level picker**
    - Description: slider for changing the cell color light level
    - Interaction:
        i. DragEnd: Update cell color to new light value and update hex color input
E. **Hex Color Input**:
    - Description: Text input for cell color. Takes hex RGB format
    - Value: Shows the value of the current cell color in RGB hex
    - Interaction:
        i. Type: If end result is valid RGB hex, update cell color and adjust color pickers to match
    - Data validation:
        i. Entered data must be valid RGB hex
F. **Randomize Cell Color Button**
    - Description: Button for randomizing cell color
    - Interaction:
        i. Hover: highlight, description tooltip
        ii. Click: Randomize cell color, and cell color inputs
G. **Background color Hue/Saturation picker**
    - Description: Wheel for picking background color hue and saturation
    - Interaction:
        i. DragEnd: Update background color to new value and update hex color input
H. **Background color light level picker**
    - Description: slider for changing the background color light level

- Interaction:
    i. DragEnd: Update background color to new light value and update hex color input

I. **Hex Color Input**:
- Description: Text input for background color. Takes hex RGB format
- Value: Shows the value of the current background color in RGB hex
- Interaction:
    i. Type: If end result is valid RGB hex, update background color and adjust background color pickers to match
- Data validation:
    i. Entered data must be valid RGB hex

J. **Randomize background Color Button**
- Description: Button for randomizing background color
- Interaction:
    i. Hover: highlight, description tooltip
    ii. Click: Randomize background color, and background color inputs

#3 Simulation Settings



A. Title

- Title of configuration group

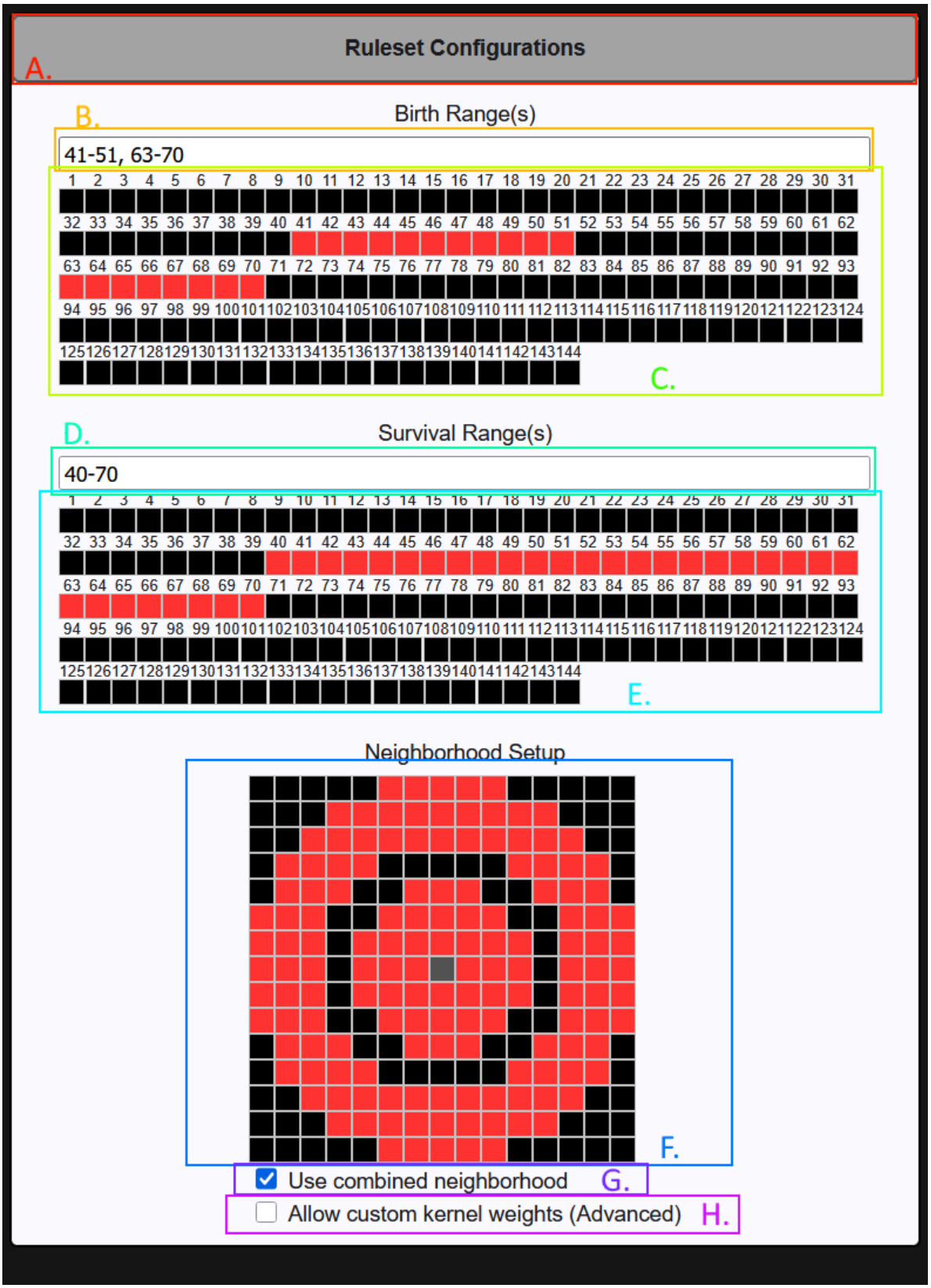
B. Frame rate slider

- Description: Slider input that controls the max simulation frame rate
- Interaction:
    - **i.** DragEnd: Update simulation max frame rate and update text representation

C. Noise Generator selector

- Description: Radio group for choosing noise generator the simulation uses
- Interaction:
    - **i.** Change: Update simulation noise generator

#4 Ruleset Configuration



A. **Title**
    - Title of configuration group
B. **Birth Range Text input**
    - Description: Text input for handling birth range rule
    - Interaction:
        i. Type: If typed range is valid, update birth range mask and update simulation with new rule
    - Data validation: Input must be a list of valid ranges, where n-m :: m > n and values are above zero, and less than or equal to sum of cells used in the birth neighborhood kernel.
C. **Birth range mask editor**
    - Description: Bitmask editor for birth rule. Array of numbered cells color to represent active ranges in the birth rule
    - Interaction:
        i. Click(index): Toggles that index of the bitmask rule, and updates the string input to match
        ii. Hover(index): Highlights hovered cell
    - Visual Changes:
        i. Activated indexes are colored red, inactive are colored black
D. **Survival Range text input**
    - Description: Text input for handling survival range rule
    - Interaction:
        i. Type: If typed range is valid, update survival range mask and update simulation with new rule
    - Data validation: Input must be a list of valid ranges, where n-m :: m > n and values are above zero, and less than or equal to sum of cells used in the survival neighborhood kernel.
E. **Survival range mask editor**
    - Description: Bitmask editor for survival rule. Array of numbered cells color to represent active ranges in the survival rule
    - Interaction:
        i. Click(index): Toggles that index of the bitmask rule, and updates the string input to match
        ii. Hover(index): Highlights hovered cell
    - Visual Changes:
        i. Activated indexes are colored red, inactive are colored black

F. Neighborhood kernel editor

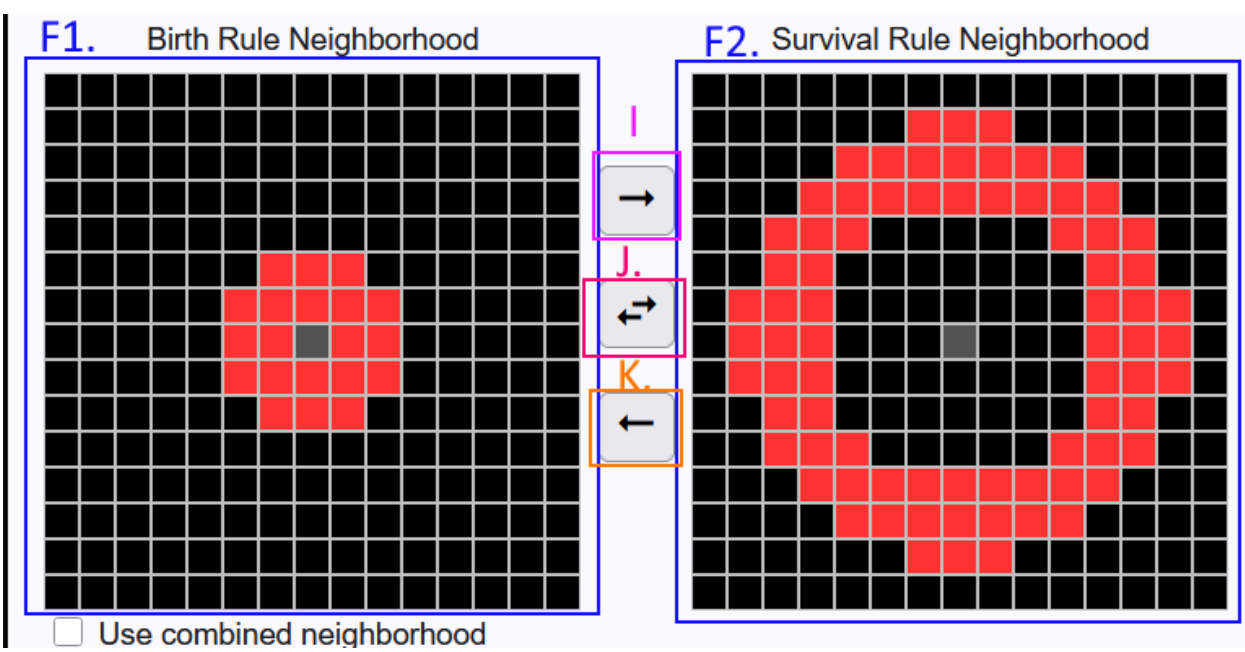
- Description: Kernel editor for combined kernel
- Interaction:
  - **i.** Click(x,y): Toggles cell value at index and updates simulation
    - **1.** When cells are toggles, update max value for rule masks
  - **ii.** Hover(x,y): Highlights hovered cell
- Visual changes:
  - **i.** Active cells are colored red, inactive cells are colored black
  - **ii.** While in combined neighborhood view, only show one nieghborhood that is used for both rules.
  - **iii.** In detailed kernel view, kernels are replace by Detailed kernel editor

G. Combined neighborhood control

- Description: Checkbox controlling whether configuration uses combined neighborhood.
- Interaction:
  - **i.** Change: Update combined neighborhood view
  - **ii.** Hover: Highlight and tooltip description

H. Custom Kernel Weight View Control

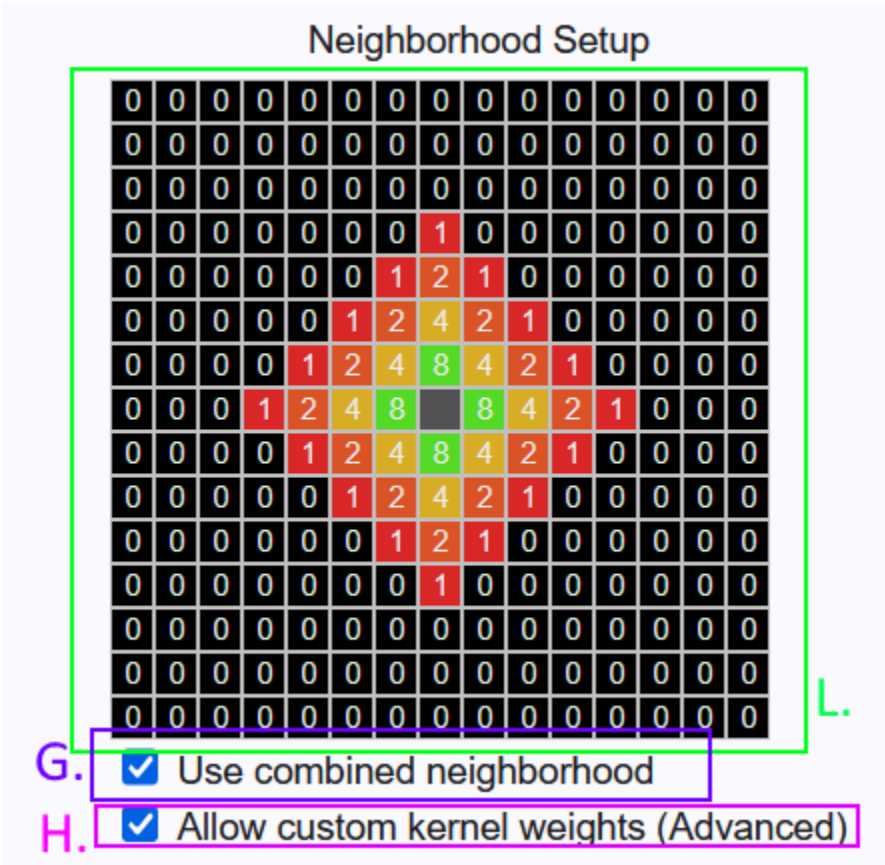
- Description: Checkbox controlling whether configuration allows custom kernel weights
- Interaction:
  - **i.** Change: update custom kernel weight view
  - **ii.** Hover: Highlight and tooltip description



I. **Copy Kernel Right Button**
   - Description: Copy the birth rule kernel onto the survival rule
   - Interaction:
     i. Click: Copy the birth rule kernel onto the survival rule
     ii. Hover: Highlight, tooltip description

J. **Swap Kernel**
   - Description: Copy the birth rule kernel onto the survival rule
   - Interaction:
     i. Click: Swap the birth kernel and survival kernel
     ii. Hover: Highlight, tooltip description

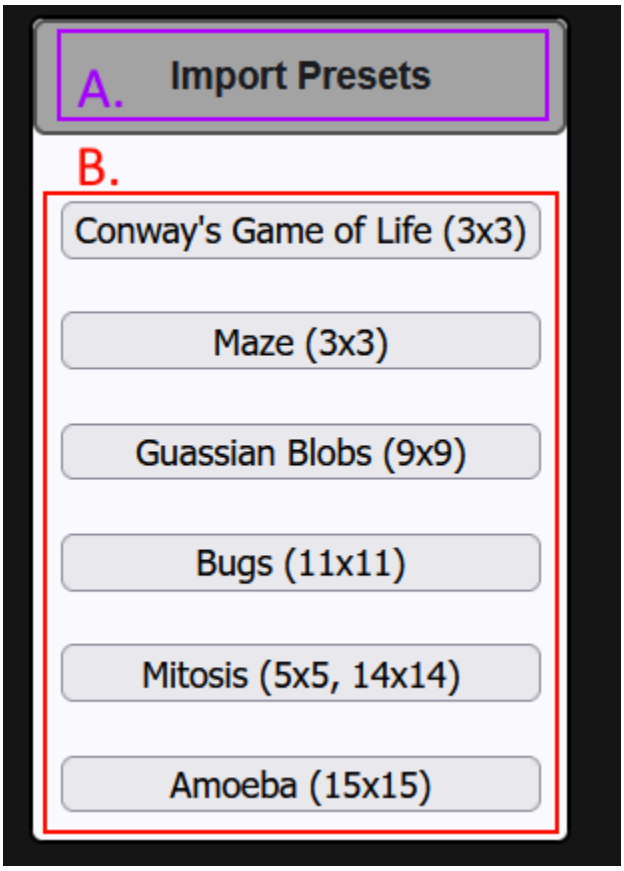
K. **Copy Kernel Left Button**
   - Description: Copy the survival rule kernel onto the birth rule
   - Interaction:
     i. Click: Copy the survival rule kernel onto the birth rule
     ii. Hover: Highlight, tooltip description



L. **Detailed Kernel Editor**
   - Description: Kernel editor used when Allow custom kernel weights is enabled

- Interaction:
    - **i.** Click(x,y): Enter typing mode to edit specific kernel cell weight
    - **ii.** Arrow keys: If in typing mode, using arrow keys will move active editing cell in direction of arrow key input
    - **iii.** Hover: Highlight cell
- Data Validation:
    - **i.** Cell inputs are only values between 0 and 9
- Visual Changes:
    - **i.** Cell color is based on a gradient, from 0 to 9
    - ii. The text value in the cell is indicated by its current kernel weight

#5 Preset Configurations



A. Title

- Title of configuration group

B. Import Option List

- Description: list of preset configuration option buttons
- Interaction:
  - **i.** Click(button): Import clicked preset option into the ruleset UI and update simulation with new configurations.
  - **ii.** Hover: Highlight option, tooltip description